QubitGuard

Neural error correction, measured against strong classical decoders.

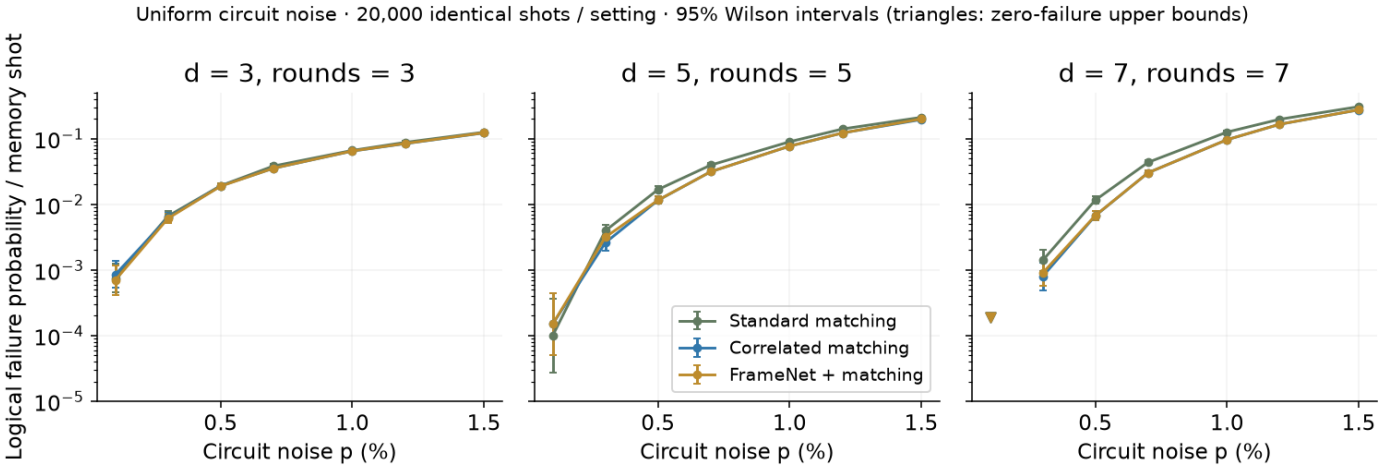
Technical report / v0.1.0 / 25 October 2026 / CPU experiments

The compact neural hybrid is genuinely trained and produces verified syndrome-consistent logical-frame corrections. It does not establish an advantage over correlated matching: 0 of 42 primary setting comparisons favor it after correction for the 364 primary sweep comparisons. 2 settings significantly favor correlated matching. The larger full-map follow-up also fails to improve the fresh test mixture. Correlated PyMatching is the recommended decoder for these measured conditions.

Independent test results at p = 0.007

Distance	Standard	Correlated	FrameNet s17
d3	763/20,000 (3.815%)	705/20,000 (3.525%)	706/20,000 (3.530%)
d5	789/20,000 (3.945%)	626/20,000 (3.130%)	626/20,000 (3.130%)
d7	869/20,000 (4.345%)	603/20,000 (3.015%)	603/20,000 (3.015%)

Uniform circuit noise; 20,000 identical test shots per method and distance. These p values were seen during training, but test streams are independent.



Evidence: 14 trained checkpoints, 1,120,000 independent primary/follow-up test shots, saved data and raw predictions. No quantum hardware, paid API or external model weights were used.

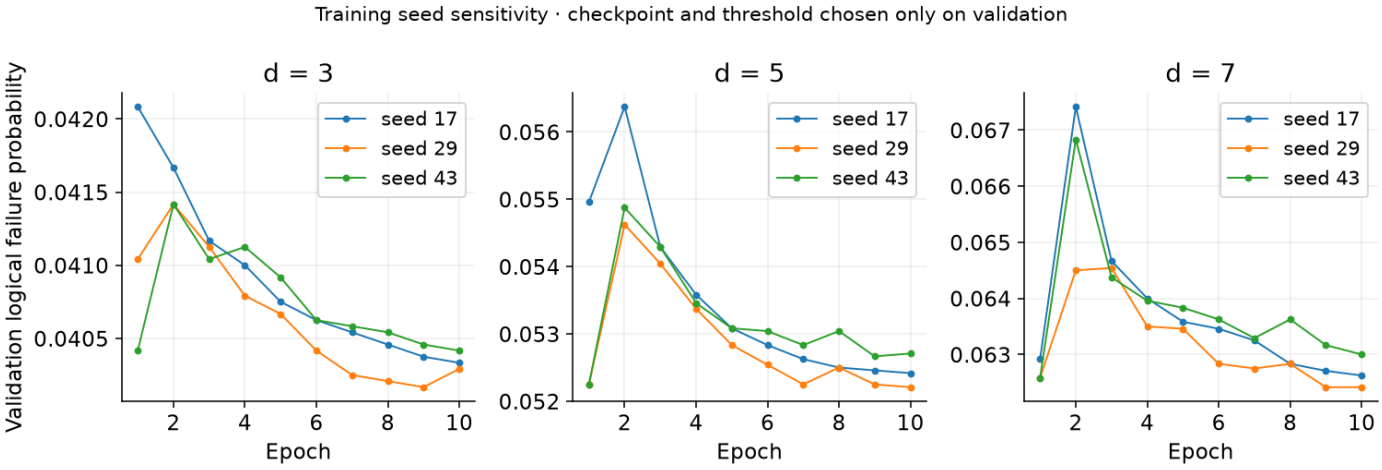
1 / What the model learns

Stim generated rotated surface-code X memories at distances 3, 5 and 7, with rounds equal to distance. Uniform noise sets after-Clifford depolarization, before-round data depolarization, before-measure flips and after-reset flips to the same nominal p. The shifted profile keeps gate p, uses idle p/4 and measurement/reset 2p. This is measurement-heavy channel imbalance, not Pauli-Z-biased noise, leakage or coherent noise. Both matching baselines receive each exact test circuit detector-error model, including decomposed correlation separators. Correlated matching is enabled in both graph construction and inference.

FrameNet reads a two-channel space-time volume: detector bits and a detector-validity mask. Width-8 spatial 1x3x3 convolutions alternate with depthwise temporal 3x1x1 convolutions. A 2x2x2 pooled readout and 48-unit head also receive standard/correlated logical predictions, their disagreement, syndrome density and declared nominal p. The 4,209-parameter model learns true logical flip XOR standard prediction. Its residual bit toggles the standard logical frame. A shortest closed odd-logical graph cycle realizes that toggle without altering any detector parity. These are graphlike DEM recoveries, not unique physical Pauli trajectories.

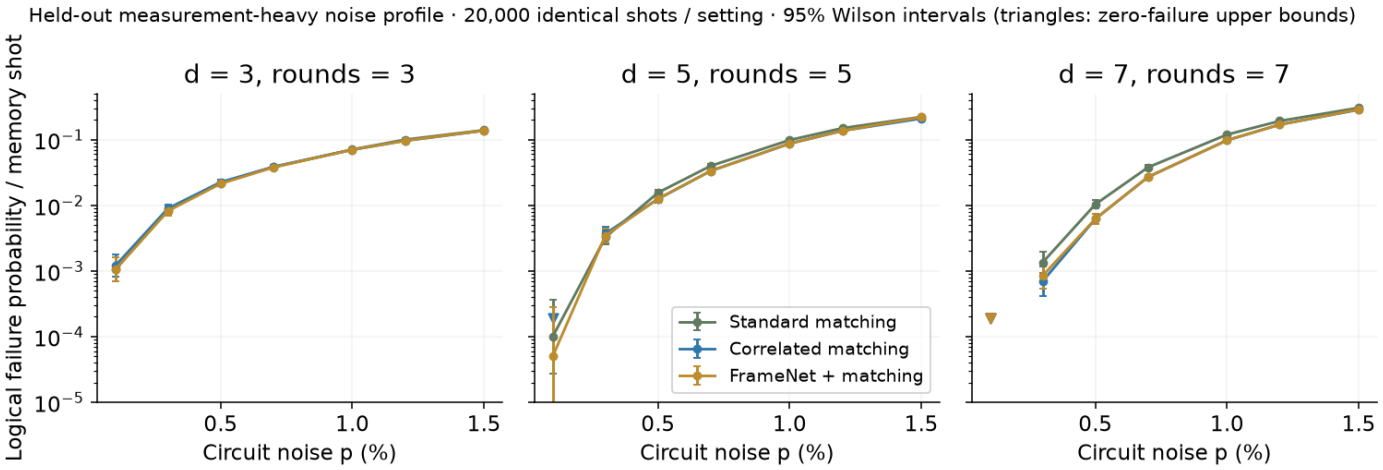
Each full compact model sees 60,000 retained training shots across p={0.003,0.007,0.012} for 10 epochs; validation uses 8,000 shots at each p. Seeds 17,29,43 vary initialization while sharing training data. AdamW uses learning rate 0.001, weight decay 0.0001, batch 512, float32 and four CPU threads. Class weighting is sqrt(negative/positive), capped at 8. Checkpoint and a grid-searched threshold are selected only on validation failures, with BCE breaking checkpoint ties. Seed17 is the declared primary model; all three initialization seeds are retained. Sigmoid values are uncalibrated class-weighted scores.

Model	Parameters	Selected epoch	Fit + validation (s)
Compact d3, seeds 17/29/43	4,209	10 / 9 / 10	32.6 - 444.2
Compact d5, seeds 17/29/43	4,209	10 / 10 / 1	60.0 - 177.7
Compact d7, seeds 17/29/43	4,209	10 / 10 / 1	324.7 - 368.2
Full-map d3, seed 17	25,713	10	176.3



Fit times include validation. Initializations share data; they are not additional independent test samples.

2 / Generalization and strong baselines



Distance / profile	Standard failures	Correlated	FrameNet s17
d3 / uniform	6866	6631	6649
d3 / measurement-heavy	7579	7479	7448
d5 / uniform	10007	8780	8878
d5 / measurement-heavy	10550	9643	9839
d7 / uniform	13705	11455	11485
d7 / measurement-heavy	13307	11716	11795

Failures per 140,000-shot equal-weight noise mixture. Each row combines seven operating points, so it is not a hardware error rate.

The primary network beats standard matching in several regimes by using correlated matching context, but this does not establish new learned accuracy beyond correlated matching. At d5 p=0.015 both profiles significantly favor correlated matching after global adjustment. A similar validation score does not guarantee robust extrapolation.

d / profile / p	Neural wins	Neural losses	Delta LER, percentage points	Adjusted p
d5 / uniform / 0.015	43	130	+0.435	8.5e-09
d5 / biased / 0.015	173	373	+1.000	2.6e-15

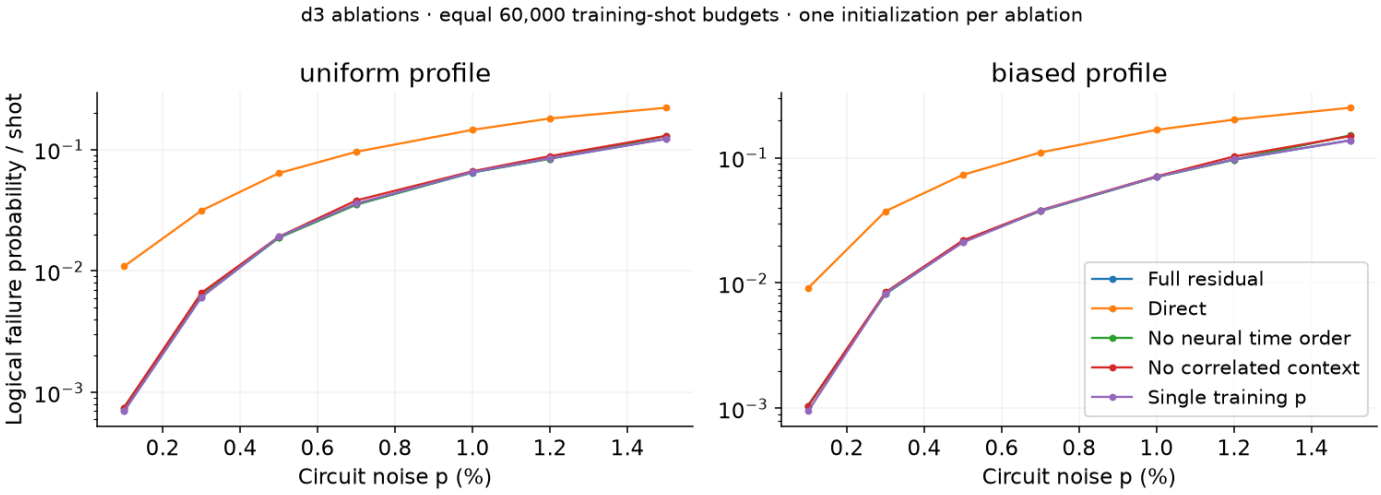
Wins: neural correct and correlated wrong. Losses: correlated correct and neural wrong. Positive delta favors correlated. Adjusted p uses all 364 primary paired comparisons.

3 / Ablations and a rejected revision

The direct head removes matching predictions and targets true logical flips. The time ablation averages neural inputs over time; matching context still uses the full history. The no-correlation control masks correlated predictions/disagreement. The single-p control uses the same total training/step budget at p=0.007 and mixed-p validation. Its seed is reused with a different sampling shape; exact prefixes differ, so it is not independent training randomness. Controls have one initialization, so small rankings are exploratory. The weak no-correlation and direct results show that much of the hybrid benefit over standard matching comes from access to correlated matching, not demonstrated new syndrome inference.

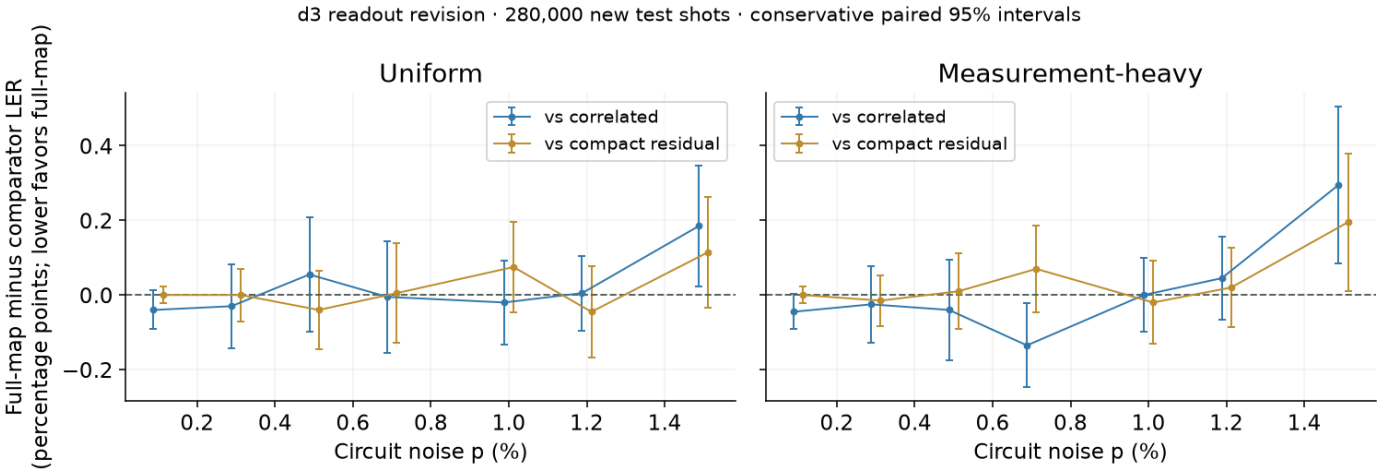
d3 model	Uniform failures	Shifted-profile failures
Standard	6866	7579
Correlated	6631	7479
Compact residual	6649	7448
Direct logical head	14973	16957
No neural time order	6796	7716
No correlated context	6974	7822
Single-p training	6688	7456

Each cell counts failures across 140,000 independent test shots over seven noise settings. Every method sees the same shots.



4 / Readout revision on fresh data

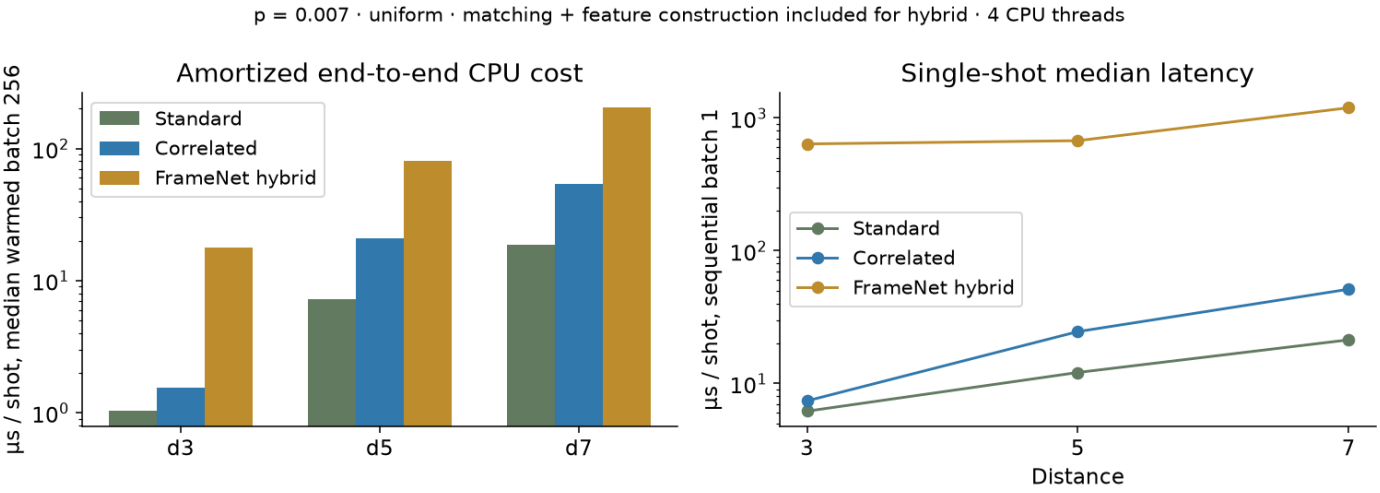
After observing the compact model mostly reproduce correlated matching, a targeted revision replaced 2x2x2 pooling with the full d3 feature map while preserving the encoder, 48-unit head, training shots, optimization steps and seed17. Parameter count rose to 25,713. It selected 971/24,000 validation failures versus compact seed17 968. A new high-bit random-stream namespace produced 280,000 independent d3 test shots across the same 14 settings. The fresh mixture is descriptive, not one physical operating point. The larger model does not improve it and is not promoted to the default demo. This adaptive follow-up is an engineering capacity test, not a preregistered scientific discovery.



Fresh d3 test / model	Uniform failures	Shifted-profile failures
Standard	6688	7615
Correlated	6495	7412
Compact residual	6503	7379
Full-map revision	6525	7431

Follow-up mixture: 140,000 shots per profile, with a new independent stream. Full-map source, checkpoint, raw predictions and circuits are retained in experiments/fullmap-v2.

5 / Full inference cost



Distance / decoder	Batch 256 us/shot	Single median us	Observed p95 us
d3 / standard	1.03	6.2	67.1
d3 / correlated	1.53	7.4	14.6
d3 / FrameNet	17.72	637.1	1378.5
d5 / standard	7.23	12.1	21.0
d5 / correlated	20.83	24.6	41.1
d5 / FrameNet	80.67	674.0	1453.0
d7 / standard	18.56	21.3	45.0
d7 / correlated	53.55	51.3	124.8
d7 / FrameNet	203.59	1192.9	3135.2

All inference costs are measured on this Apple arm64 Mac using four PyTorch CPU threads, float32. Batch throughput is median warmed wall time of five batches of 256, divided by 256. Sequential single-shot median and empirical p95 use 32 shots. Hybrid costs include both matching calls, features, grid allocation/tensor conversion and neural inference. Sampling, disk, HTTP, graph construction and certificate construction are excluded. Final timings were refreshed without concurrent QubitGuard fits/tests; OS background load is uncontrolled. The 32-shot tail statistic is not a real-time bound. Fit times include validation but exclude prior data generation/baseline precomputation, and are observational under variable machine load, not a causal architecture speed ranking.

The hybrid incurs both matching calls and neural overhead. Its measured cost and accuracy do not justify recommending it over correlated matching here. The larger readout is also retained as an unsuccessful engineering revision, rather than advertised as an improvement.

Environment: Python 3.13.2; PyTorch 2.14.1; Stim 1.16.0; PyMatching 2.4.0; arm64.

6 / Statistical and reproducibility protocol

Every primary setting has 20,000 independent held-out shots; all methods decode exactly the same retained vectors. LER is failures per full memory shot, not a fitted per-round rate. Marginal intervals are 95% Wilson. Paired wins/losses identify discordant errors. The exact two-sided paired binomial test is equivalent to exact McNemar on discordances. Global Bonferroni adjustment covers 364 primary comparisons. A conservative paired difference interval subtracts simultaneous Clopper-Pearson win/loss probability bounds, retaining nonzero uncertainty with zero disagreements. These intervals are marginal; simultaneous claims use adjusted p. Absence of significance does not establish equivalence.

The delivery retains exact circuits/DEMs, 61 primary NPZ datasets (1,152,000 stored shots including training/validation/control), 14 follow-up test datasets, 14 trained checkpoints, raw held-out predictions, checksums, exact dependency versions and source/config compatibility fingerprints. A full d3 seed17 retraining reproduced every learned tensor, threshold and selected epoch; its recorded checkpoint bytes also matched. Later cache/atomic-storage safeguards do not alter the numerical fitting rule. Initial artifact compatibility metadata was added transparently after full exact circuit, sample, baseline and feature replay. That audit is compatibility evidence, not a retroactive claim of immutable execution provenance. Exact cross-platform replay is not guaranteed.

The initial combined runner stopped before evaluation when a new guard rejected an obsolete ignored split-seed field loaded at process startup. The effective numerical training configuration was unchanged. Current strict CLI execution validates complete checkpoint pairs, relevant source/config/dependency signatures, dataset identities and retained circuit bytes before reuse. Incomplete artifacts are preserved and rejected. This history is recorded instead of implying an uninterrupted pristine run.

Automated tests reconstruct sampled DEM syndrome/logical truth and displayed correction parity without trusting returned certificate booleans. They also alter hidden truth to verify that inference cannot see it, compare an independent uncorrelated graph, exercise concurrent requests, and reject metadata/source/data mutations. Final test execution is recorded with the release artifacts.

Reproduce with: qubitguard verify; python scripts/replay_datasets.py --root .; qubitguard evaluate; qubitguard latency; qubitguard plot. New fitting experiments require a fresh root and the versioned configuration. See README for exact setup and model cards for supported conditions.

The local server has no external account/API dependency. Checkpoint integrity is checked before loading, shared mutable matching workspaces are serialized, unsupported neural scope is explicit, and a failed request preserves the prior valid shot.

7 / Interactive system and research boundaries



Current-source live playground, captured from the real local simulator. This shot is illustrative and is not benchmark evidence.

The playground shows a surface-code lattice and space-time detector history with stepping/playback, noise and distance controls, sampled error mechanisms, neural predictions, graph corrections and logical success/failure. It distinguishes a valid syndrome recovery from the correct logical class. Representative physical locations are explanatory possibilities, not a unique observed trajectory.

No result here establishes a surface-code threshold, universal logical-operation coverage, neural distance transfer, quantum-hardware accuracy, leakage robustness, analog-readout decoding, streaming operation or real-time capability. Neural models support X memory only; the simulator also runs Z memories with explicit unavailable neural output. Unseen noise tests cover selected interpolation/extrapolation and channel imbalance only. The d7 seed29 shifted p=0.015 result is particularly unstable (6,338 failures versus correlated 5,703), despite similar validation performance. Stronger baselines, richer local representations and substantially larger rare-error training sets remain research directions, not proven remedies.

Published NVIDIA GB300 timing and AlphaQubit hardware results are research context, not local reproductions. Their noise, training scale and hardware differ. NVIDIA source and model-weight licenses are separate; gated weights were not downloaded or redistributed. Newly initialized QubitGuard weights and original synthetic data use the project MIT license, while dependencies retain their licenses.

Primary foundations: Bausch et al., Nature 635, 834-840 (2024), doi:10.1038/s41586-024-08148-8; Chamberland et al., arXiv:2604.12841v1 (2026); Gidney, Quantum 5,497 (2021); Higgott and Gidney, Quantum 9,1600 (2025); Fowler, arXiv:1310.0863; Meinerz et al., arXiv:2101.07285v2; Gicev et al., arXiv:2506.16113; Lange et al., arXiv:2610.03869. See research-foundations.md and references.bib for links, prior-art boundaries and separate source/model licenses.

Original engineering: independently implemented simulation/data pipeline, compact logical-frame model, parity-certified recovery construction, strict paired protocol, readout-capacity follow-up, integrity checks and real simulator UI. No claim of a novel decoder family or groundbreaking superiority is made.